

What Does a Heart Sound Classifier Actually Learn? A Controlled Comparison of Cepstral, Spectral and Wavelet-Packet Representations with Cardiac-Cycle-Aligned Explanations

Ouardi Ilyass

Dipartimento di Informatica
Università degli Studi di Milano
Milan, Italy

ilyass.ouardi@studenti.unimi.it

Abstract—Automatic screening of phonocardiogram (PCG) recordings could extend cardiac auscultation to settings without expert cardiology, but on the PhysioNet/CinC 2016 corpus a high aggregate score is easy to obtain and hard to trust, because the recording site is itself strongly predictive of the diagnosis. This project compares three feature representations (MFCC, log-Mel spectrogram, perceptual wavelet packet) and three classifiers (SVM, random forest, small CNN) under one recording-level protocol, and pairs each performance claim with a diagnostic that could refute it: unsupervised k -means structure, a per-site breakdown, and a falsifiable test of whether attributions concentrate in systole. The best model reaches a modified accuracy of 0.866, comparable to published entries, but k -means recovers the acquisition site far better than the diagnosis, and per-site modified accuracy ranges from 0.99 down to 0.48. A direct test confirms the mechanism: the recording site is recoverable from the same features at 0.65 balanced accuracy (versus 0.17 chance), and diagnostic performance falls from 0.84 to 0.46 — essentially chance — under leave-one-site-out evaluation, though a weak signal survives once site is held fixed. Attribution is only weakly systole-enriched, and no more so for correct than for incorrect predictions. We conclude that the aggregate metric substantially overstates what the models have learned.

I. INTRODUCTION

Cardiac auscultation is the oldest and cheapest cardiac screening tool in clinical use, but interpreting heart sounds by ear is difficult and depends heavily on the listener’s training [1], [2]. The two dominant events, the first and second heart sounds (S1 and S2), are short transients carrying most of their energy below 100 Hz; murmurs are noise-like signals occupying the intervals between them and extending to several hundred hertz [3]. Automatic analysis of the phonocardiogram (PCG) is attractive wherever expert cardiology is scarce: primary-care triage, telemedicine, and low-cost digital stethoscopes. The task addressed here is the binary one posed by the PhysioNet/Computing in Cardiology (CinC) 2016 Challenge — label a recording *normal* or *abnormal*, where abnormal denotes a confirmed cardiac pathology rather than merely an audible murmur [4], [5].

Work on this corpus is well developed. The winning entry combined a feature-based classifier ensemble with a convo-

lutional network on band-limited cardiac cycles [6]; a purely convolutional system on MFCC heat maps followed [7], as did feature ensembles that avoid segmentation altogether [8]. Wavelet-packet descriptions of the PCG form a separate line [9], [10], motivated by the fact that the informative events are transients of tens of milliseconds, for which the local-stationarity assumption behind the short-time Fourier transform fits poorly [11]. Reported modified accuracies cluster around 0.84–0.86.

Less developed is the question of *what* such a system learns. The public portion of CinC 2016 comprises six sub-databases collected by different groups with different equipment, and abnormal prevalence varies across them by roughly 69 pp. Site identity is therefore strongly predictive of the label, and a model that recognises the recording device or the room can score well without learning anything about cardiac acoustics — the failure mode documented for radiographic COVID-19 detectors [12]. Here it is not hypothetical; it is the dominant structure in the data.

This project therefore treats the aggregate metric as a claim to be tested. We implement one recording-level pipeline and use it to compare three representations that differ in kind — cepstral (MFCC), spectral (log-Mel) and perceptual wavelet packet (PWP) — across three classifier families. Each performance claim is paired with a diagnostic that could refute it. k -means asks whether the diagnosis is geometrically present in each feature space *before any label is used*, with cluster agreement measured against both the diagnosis and the recording site so that a null result is still informative. Performance is broken down per sub-database, since a single bright site exposes a shortcut. And the explainability analysis is framed as a falsifiable prediction: because most clinically important adult murmurs are systolic, a detector that has learned pathology should place more attribution mass in systole than uniform temporal attention would, which can be tested against an analytic and a permutation null.

The scope is narrow. We propose no new architecture and make no claim beyond the split we evaluate; the official

challenge test set was never released, so all comparison with published entries is contextual. The contribution is the controlled comparison and the diagnostics — and two of the three expectations we began with were contradicted by the evidence. They are reported here in the same voice as the results that held.

II. METHOD

Fig. 1 shows the pipeline. Its distinguishing property is not any single block but the fact that the three feature domains are computed from *identical* windows and evaluated under a single recording-level protocol, so that any difference between them is attributable to the representation and not to the experimental setup. The remainder of this section describes each stage and the reasoning behind the choices that are not self-evident.

A. Preprocessing

Recordings are resampled to 2 kHz and band-pass filtered to 25–400 Hz with a fourth-order Butterworth design. Below 25 Hz the signal is respiration, movement and baseline drift; above 400 Hz there is essentially no cardiac acoustic energy. The band covers S1 (≈ 25 –45 Hz), S2 (≈ 55 –75 Hz) and murmur energy.

Filtering is applied forward and backward, so that it is zero-phase. This matters specifically for the analysis in Section VI: a causal filter has frequency-dependent group delay, which would shift high-frequency murmur energy relative to the low-frequency S1 fundamental, and the alignment test is precisely a claim about *when* evidence occurs.

Friction transients from the stethoscope diaphragm are suppressed by the spike removal procedure of Schmidt et al. [13]; these are the loudest events in many recordings and would otherwise dominate every energy-based descriptor. Finally each recording is z -normalised. Gain is a site artifact — different clinics used different preamplifiers — and since gain correlates with site and site correlates with prevalence, preserving absolute level would hand the model a shortcut.

B. Windowing

Windows are 3 s long and fixed. At 60–100 bpm a cardiac cycle lasts 0.6–1.0 s, so a 3 s window always contains at least three complete cycles and is robust to the phase at which it happens to start. Training uses 50 % overlap as augmentation; validation and test use non-overlapping windows, so that evaluation samples remain statistically independent and confidence intervals are not artificially narrowed.

Windows are deliberately *not* cut at cardiac-cycle boundaries. Doing so would build the location of systole into the input representation and would make the alignment claim of Section VI circular.

C. Three feature representations

The three representations were chosen to differ in kind rather than in parameter setting.

MFCC. Thirteen cepstral coefficients from a mel filterbank, with first and second time derivatives, each trajectory summarised over the window by six statistics (mean, standard

deviation, minimum, maximum, skewness, kurtosis), giving 234 dimensions. This is the standard baseline in the CinC 2016 literature [7], [14], so it anchors comparability, and the cepstrum is a decorrelated description of spectral shape, which is broadly what separates murmur from no-murmur.

Log-Mel spectrogram. The same filterbank output retained as a two-dimensional time–frequency map and used as CNN input. This is the only representation that preserves the time axis; the statistical pooling used for the classical branch discards it. The contrast between the two is one of the comparisons the study is designed to make.

Perceptual wavelet packet (PWP). A Daubechies-4 wavelet packet decomposition [15] whose uniform subbands are merged into perceptually spaced bands, each described by log energy, relative energy, Shannon entropy, standard deviation, skewness, kurtosis and peak amplitude, following the construction of Safara et al. [9]. MFCC and log-Mel are both STFT-based and assume local stationarity within each window, which fits poorly a signal whose informative events are 10–50 ms transients; the packet transform tiles the time–frequency plane adaptively [10], [11]. A further property matters in Section VI: PWP features map onto frequency bands exactly, with no transform in between.

Two resolution constraints shaped these parameters and are worth stating because they otherwise look arbitrary. First, the number of mel bands is limited by FFT resolution: at 2 kHz with a 256-point FFT the bin width is 7.81 Hz, so only about 48 usable bins fall inside 25–400 Hz, and requesting more filters than that produces empty, all-zero filters that inject structural zeros into every feature vector. Second, wavelet packet nodes are not returned in frequency order: the packet tree reverses the frequency axis in the high-pass branch at each level, so the natural ordering is a bit-reversed permutation of frequency order, and indexing it as though it were frequency-ordered silently scrambles the spectral axis without raising an error. Nodes are reordered explicitly before grouping.

D. Classifiers

Three families with different inductive biases are compared: an RBF-kernel SVM (global, distance-based, maximum-margin); a random forest (local, axis-aligned, ensemble, and additionally admitting an exact SHAP computation [16]); and a small two-dimensional CNN on log-Mel input, comprising four Conv–BN–ReLU–MaxPool blocks, global average pooling and a small fully connected head, on the order of 10^5 parameters. The network is kept deliberately small because the dataset is small; a large pretrained network would memorise it and would obscure the feature-domain comparison that is the point of the study.

Class imbalance is handled by inverse-frequency weighting in all three families. Augmentation uses circular time shift — the phase at which a window starts is arbitrary, so the label must be invariant to it — and additive Gaussian noise. Frequency masking is deliberately avoided: masking a band could delete the murmur itself, converting an abnormal example into a mislabelled normal one.

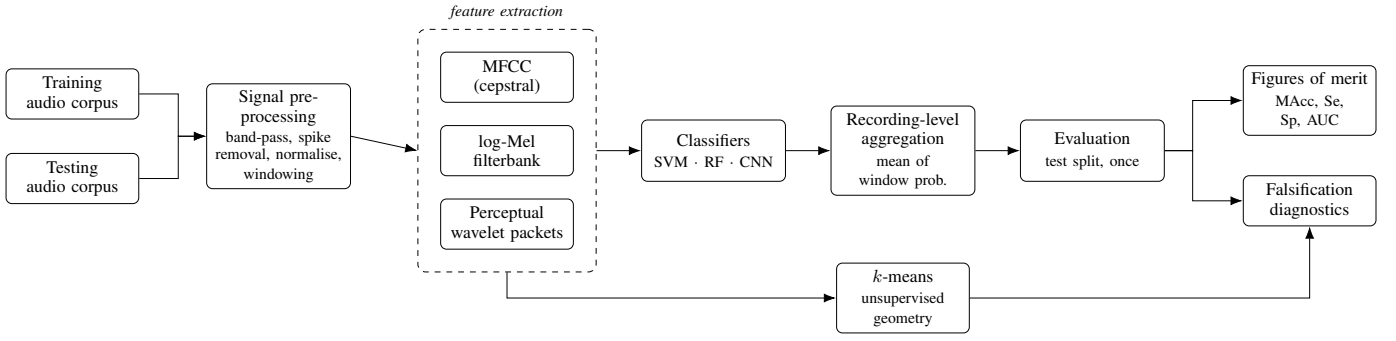


Fig. 1. Block diagram of the classification system. Training and testing recordings share one preprocessing chain and are mapped into three feature representations that differ in kind (dashed group); a single recording-level protocol then classifies, aggregates window scores to a recording decision, and evaluates. The lower branch feeds the unsupervised geometry into the diagnostics that test what the aggregate metric conceals. Each stage is expanded in Figs. 2–3.

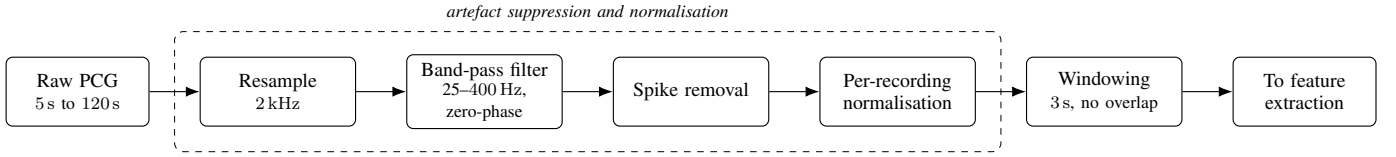


Fig. 2. Preprocessing. Recordings are resampled to 2 kHz, band-pass filtered to the diagnostic 25–400 Hz range with a zero-phase (forward–backward) filter that preserves event timing, despiked, amplitude normalised per recording, and cut into non-overlapping 3 s windows. The same windows feed all three feature domains.

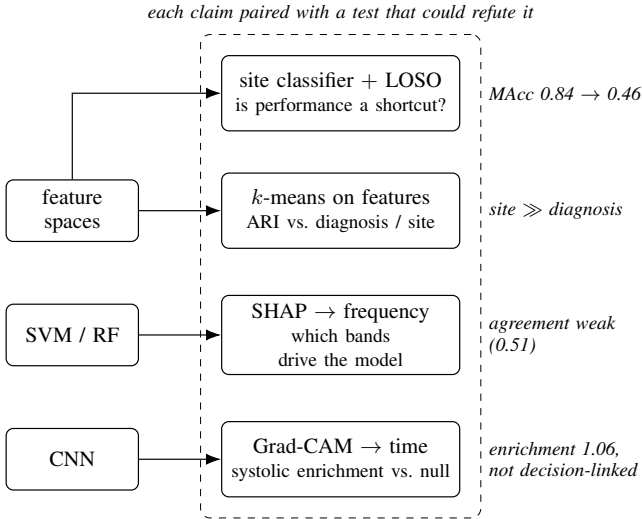


Fig. 3. Falsification diagnostics. Each performance claim is paired with a test capable of refuting it: unsupervised k -means geometry, SHAP frequency attribution, Grad-CAM temporal enrichment against analytic and permutation nulls, and a site classifier with leave-one-site-out re-evaluation. The last confirms that the pooled MAcc largely reflects the recording site rather than transferable cardiac acoustics.

Window scores are aggregated to a recording decision by averaging predicted probabilities, because a clinician diagnoses a patient rather than a three-second excerpt. Majority voting was also computed and gave results within 0.004 MAcc of probability averaging.

III. EXPERIMENTAL SETUP

A. Dataset

We use the public training portion of the PhysioNet/CinC 2016 corpus [4], [17]: 3240 mono recordings, 20.2 h in total, sampled at 2 kHz, with durations from about 5 s to 120 s. Labels are per recording, and 20.5 % of recordings are abnormal, an imbalance ratio of 3.87:1. The corpus is divided into six sub-databases, `training-a` through `training-f`, each collected by a different group at a different site with different equipment. Abnormal prevalence differs across them by 68.9 pp, which is the single most important property of this corpus for the present study.

All results below therefore come from a self-defined split of the six subdataset, and published challenge scores are context rather than a like-for-like comparison. We restate this wherever those numbers appear.

B. Splits

Splits are made at the *recording* level, never at the window level, and stratified jointly on label and sub-database. Recording-level splitting prevents windows of the same patient — sharing a stethoscope, a noise floor and a murmur — from appearing in more than one partition; this is the most consequential guard in the whole protocol. Joint stratification keeps both the class balance and the site mix constant, so that the test partition cannot end up dominated by one clinic. The realised deviation is at most 0.07 pp in class balance and 0.24 pp in site composition.

Windowing yields 30 695 training, 3329 validation and 3362 test windows, the latter drawn from 484 recordings of which

TABLE I

UNSUPERVISED STRUCTURE OF EACH FEATURE SPACE AT $k = 2$, ON THE TRAINING SPLIT. ARI AGAINST THE DIAGNOSIS IS COMPARED WITH ARI AGAINST THE RECORDING SITE TO IDENTIFY WHAT THE CLUSTERS ACTUALLY TRACK.

Features	Silh.	DB	ARI (class)	ARI (site)	PCA dims (95%)
MFCC	0.396	1.512	0.018	0.201	51
log-Mel	0.192	1.703	0.028	0.043	37
PWP	0.388	1.218	0.002	0.159	31

20.5 % are abnormal. No recording was excluded for quality or loading failure.

C. Parameters

Features use a 256-point Hann window with hop 32 (16 ms), 32 mel bands over 25–400 Hz. MFCC uses 13 coefficients with Δ and Δ^2 (width 5) and six aggregation statistics, giving 234 dimensions; log-Mel windows are 32×188 ; PWP uses a level-6 db4 tree grouped into 12 perceptual bands with 7 descriptors each, giving 84 dimensions. All classical features are standardised with statistics fitted on the training split only. No degenerate all-zero mel band and no near-constant feature was produced.

Classical hyperparameters were chosen by five-fold stratified group cross validation on the training split, grouping by recording and scoring balanced accuracy. The selected SVM uses $C = 10$, $\gamma = 5 \times 10^{-4}$, with cross-validated MAcc 0.874 ± 0.008 . The CNN was trained for at most 60 epochs with Adam at 10^{-3} , batch size 64, dropout 0.3, plateau scheduling and early stopping on validation MAcc; it stopped after 20 epochs with its best epoch at 8 and validation MAcc 0.915. Model selection, hyperparameter search and decision thresholds used training and validation data only; the test partition was read once, after selection was frozen.

D. Metrics and statistics

The primary metric is the official challenge metric, modified accuracy $\text{MAcc} = (\text{Se} + \text{Sp})/2$, which is identical to balanced accuracy. Plain accuracy is reported but is not an appropriate headline: with roughly 80 % normal recordings, a model that always predicts *normal* scores 0.795 accuracy at 0.500 MAcc. Trivial baselines appear in Table II so that the reader need not do this arithmetic. Sensitivity, specificity, precision, F1, ROC-AUC and the Matthews correlation coefficient are also reported.

Confidence intervals are obtained by bootstrap over *recordings*, which is the correct unit of independence; resampling windows would treat the several windows of one recording as independent observations and would produce intervals far too narrow. Pairwise model comparison uses McNemar’s test [18], because all models are evaluated on the same recordings and the samples are therefore paired, with Holm–Bonferroni correction across the family of ten comparisons.

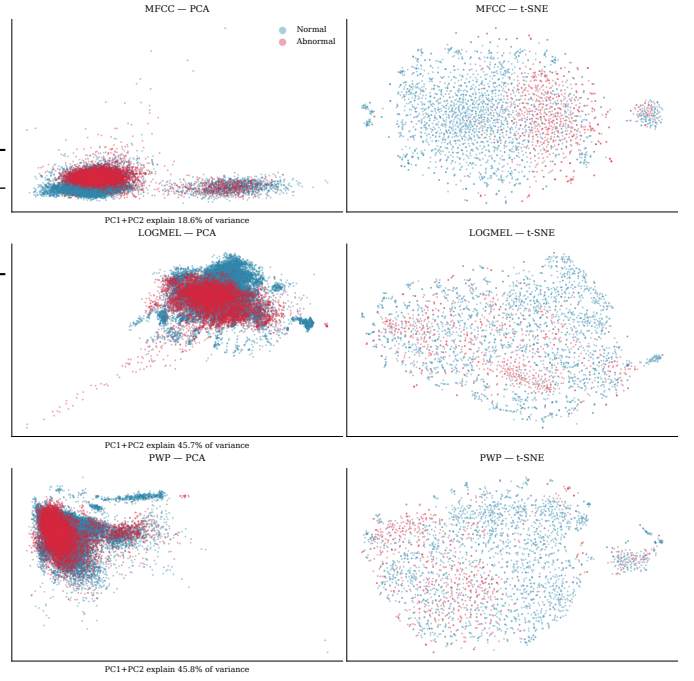


Fig. 4. PCA and t-SNE projections of the three feature spaces, one row per feature domain, coloured by diagnosis. No projection separates the two classes; the first two principal components explain only 18.6 % of the variance in the MFCC space.

IV. RESULTS I: WHAT IS IN THE FEATURE SPACE BEFORE SUPERVISION?

k -means is used here as a diagnostic rather than as a decorative preliminary. It answers a question the classifiers cannot: is the normal/abnormal distinction geometrically present in each feature space *before any label is used*? We sweep $k \in \{2, \dots, 8\}$ on the training split, and measure agreement of the partition against two references — the diagnosis and the recording site — so that a null result against the diagnosis is still informative.

Table I reports the $k = 2$ result. The adjusted Rand index against the diagnosis is 0.018 for MFCC, 0.028 for log-Mel and 0.002 for PWP: all three are close to chance. Against the recording site the same partitions score 0.201, 0.043 and 0.159. For MFCC and PWP the site agreement exceeds the class agreement by an order of magnitude. The same pattern holds at other k : for MFCC at $k = 3$, ARI reaches 0.271 against the diagnosis but 0.420 against the site.

Fig. 4 shows the corresponding PCA and t-SNE projections, in which no domain visibly separates the two classes. We stress that the site/diagnosis asymmetry is established by the index values above and not by these plots: the projections are coloured by diagnosis only, and in any case t-SNE cluster sizes and inter-cluster distances are not quantitatively meaningful, so a projection is at best corroborating evidence for a partition statistic. The comparison of ARI against two references is the actual measurement.

Two observations follow. First, the expectation we started from — that the diagnosis would be recoverable without labels

TABLE II

TRIVIAL BASELINES ON THE TEST SET (484 RECORDINGS). THE ALWAYS-NORMAL PREDICTOR ILLUSTRATES WHY ACCURACY IS NOT AN APPROPRIATE HEADLINE METRIC HERE.

Baseline	Acc.	Se.	Sp.	MAcc
always normal	0.795	0.000	1.000	0.500
always abnormal	0.205	1.000	0.000	0.500
random stratified	0.638	0.212	0.748	0.480

TABLE III

RECORDING-LEVEL PERFORMANCE ON THE HELD-OUT TEST SET, ORDERED BY MAcc. MAcc = (Se + Sp)/2 IS THE PRIMARY METRIC; 95 % BOOTSTRAP CONFIDENCE INTERVALS ARE RESAMPLED OVER RECORDINGS. ALL FIVE INTERVALS OVERLAP.

Model	Acc.	Se.	Sp.	MAcc	95 % CI	F1	AUC
SVM-MFCC	0.816	0.949	0.782	0.866	[0.834, 0.894]	0.679	0.922
CNN-log-Mel	0.851	0.889	0.842	0.865	[0.828, 0.900]	0.710	0.961
RF-MFCC	0.802	0.949	0.764	0.857	[0.824, 0.886]	0.662	0.906
SVM-PWP	0.785	0.909	0.753	0.831	[0.794, 0.863]	0.634	0.877
RF-PWP	0.731	0.970	0.670	0.820	[0.790, 0.847]	0.596	0.864

— is contradicted. Second, and more usefully, the null result is not uninformative, because the site comparison identifies what the dominant variance actually encodes. The class boundary in these spaces is a thin supervised direction inside a geometry organised primarily by how and where the recording was made. This is the first of three independent indications pointing the same way, and it is the reason we treat the aggregate metric of Section V with caution rather than reporting it as a headline.

The moderate silhouette values (0.396 for MFCC, 0.388 for PWP) should not be read as evidence of good clustering: at $k = 2$ the MFCC partition is extremely unbalanced (29 089 versus 1606 windows), and a silhouette computed on one large and one small compact cluster is easy to obtain. Davies–Bouldin and Calinski–Harabasz degrade sharply for $k \geq 3$ in every domain, which is consistent with there being no strong natural partition at any k we tested.

V. RESULTS II: CLASSIFICATION

A. Aggregate performance

Table III gives recording-level test performance for the five feature/model pairings, and Table II the trivial baselines that put them in perspective. All five models clearly exceed the always-normal predictor: MAcc ranges from 0.820 to 0.866 against 0.500. The Matthews correlation coefficient, which is zero for any constant predictor, is between 0.59 and 0.66 for the leading models. This claim is supported without qualification.

The claim that the CNN would outperform the best classical pairing is not. SVM-MFCC reaches MAcc 0.866 95 % CI [0.834, 0.894] and the CNN 0.865 [0.828, 0.900], a difference of 0.0004 inside intervals that almost entirely coincide. On MAcc the two are statistically indistinguishable, and so, on the interval criterion, are all five models.

The confusion matrices (Fig. 5) show what the aggregate hides: the models differ chiefly in where they sit on the

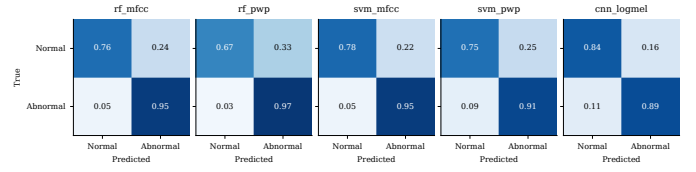


Fig. 5. Recording-level confusion matrices. The models differ mainly in where they sit on the sensitivity/specificity trade-off rather than in overall discrimination.

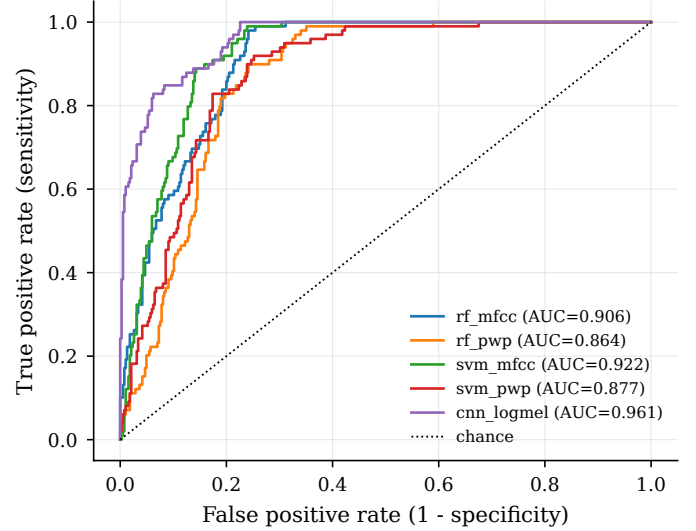


Fig. 6. ROC curves on the test recordings. The CNN attains the highest AUC (0.961) while not attaining the highest MAcc, indicating that its ranking of recordings is better than its operating point.

sensitivity/specificity trade-off. RF-PWP achieves the highest sensitivity of all (0.970) at the cost of the lowest specificity (0.670); the CNN is the most balanced (0.889/0.842). For a screening application, where a missed pathology costs more than a referral, the high-sensitivity operating points are arguably preferable, but nothing in our evaluation selects among them. The ROC curves (Fig. 6) add a further nuance: the CNN has the best AUC (0.961 against 0.922 for SVM-MFCC) without the best MAcc, meaning its *ranking* of recordings is better than its chosen operating point. A threshold tuned on validation data captures only part of that advantage.

B. Paired comparison

MAcc intervals and paired error tests do not agree, and the disagreement is informative rather than contradictory. Table IV shows that after Holm–Bonferroni correction the CNN makes significantly fewer recording-level errors than every other model, including SVM-MFCC (28/11 discordant, $p = 1.0 \times 10^{-2}$), while the two MFCC classical models are indistinguishable from each other ($p = 0.21$).

These statements are compatible. McNemar tests the count of *discordant errors* at each model’s fixed operating point; MAcc averages sensitivity and specificity, which weights the minority class far more heavily. The CNN gets more recordings right overall, but its errors are distributed such that the

TABLE IV

PAIRWISE McNEMAR TESTS ON RECORDING-LEVEL PREDICTIONS. DISCORDANT COUNTS ARE (A CORRECT / B CORRECT). SIGNIFICANCE IS ASSESSED AFTER HOLM–BONFERRONI CORRECTION ACROSS THE FAMILY OF TEN COMPARISONS.

Comparison	Discordant	p	Sig.
CNN–log–Mel vs. RF–PWP	68/10	1.1×10^{-10}	✓
RF–PWP vs. SVM–MFCC	8/49	1.2×10^{-7}	✓
RF–MFCC vs. RF–PWP	39/5	6.5×10^{-7}	✓
CNN–log–Mel vs. SVM–PWP	43/11	2.5×10^{-5}	✓
RF–PWP vs. SVM–PWP	9/35	1.6×10^{-4}	✓
CNN–log–Mel vs. RF–MFCC	35/11	7.0×10^{-4}	✓
CNN–log–Mel vs. SVM–MFCC	28/11	1.0×10^{-2}	✓
SVM–MFCC vs. SVM–PWP	27/12	2.5×10^{-2}	–
RF–MFCC vs. SVM–MFCC	8/15	2.1×10^{-1}	–
RF–MFCC vs. SVM–PWP	24/16	2.7×10^{-1}	–

TABLE V

BEST MODEL (SVM–MFCC) BROKEN DOWN BY ACQUISITION SUB-DATABASE. PERFORMANCE IS NOT UNIFORM: MAcc SPANS 0.509. TRAINING–C ($n = 4$) IS OMITTED AS IT CONTAINS A SINGLE CLASS IN THE TEST SPLIT.

Sub-database	n	Prev.	MAcc	Se.	Sp.
training-a	61	0.721	0.477	0.955	0.000
training-b	73	0.205	0.511	0.867	0.155
training-d	8	0.500	0.500	1.000	0.000
training-e	321	0.087	0.986	1.000	0.973
training-f	17	0.294	0.692	0.800	0.583

sensitivity/specificity average is unchanged. Since MAcc is the metric this task is defined by, we do not claim the CNN is the better model; we state that it is better on total errors and equivalent on the primary metric.

The feature domain matters more than the classifier family. Both MFCC models outrank both PWP models, and every MFCC-versus-PWP comparison that is significant favours MFCC. The wavelet-packet representation, adopted on the principled grounds of Section II, did not pay off here. We note in Section VI that it nevertheless remains the more useful representation for interpretation, which is a different property from accuracy.

C. The aggregate metric is not uniform across sites

Table V and Fig. 7 decompose the best model’s performance by acquisition sub-database, and this is the result that most changes how the preceding numbers should be read. Performance is not uniform. MAcc is 0.986 on training-e and 0.477 on training-a, a spread of 0.509; on training-a, training-b and training-d the model is at or below the 0.500 of a constant predictor. Specificity collapses to 0.000 on training-a and 0.155 on training-b: on those sites the model labels almost everything abnormal.

The pooled figure is therefore an average dominated by one site. training-e contributes 321 of the 484 test recordings (66%) and has a low abnormal prevalence of 8.7%; it is both the largest and the easiest partition. training-a, by contrast, has 72.1% abnormal prevalence, and a model that

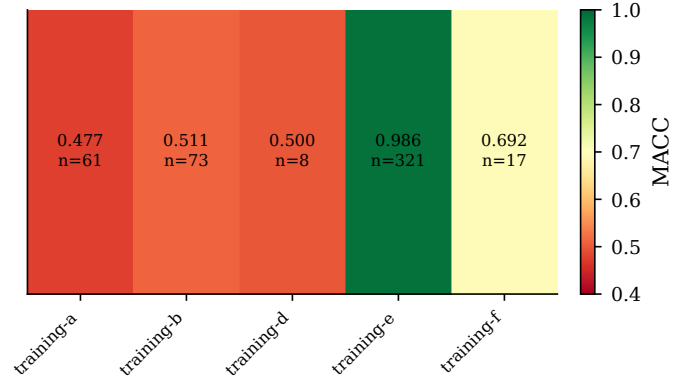


Fig. 7. Per-sub-database MAcc of the best model, with sub-database size. The pooled figure of 0.866 is dominated by training-e, which supplies 66% of the test recordings and on which the model is nearly perfect.

answers “abnormal” there is right 72% of the time by accident of prevalence.

This is exactly the behaviour Section IV predicted from the unsupervised geometry, and it is what a site-prevalence shortcut looks like when measured. We are careful about the inference: the per-site breakdown establishes that performance does not transfer across acquisition conditions, which is a measurement. That the mechanism is prevalence-driven site recognition is a further inference, supported by the clustering result and by the direction of the failures. The natural confirming experiment — training a classifier to predict the sub-database from the same features, and re-evaluating the diagnosis across sites rather than pooled — is reported next in Section V-D.

The smaller sub-databases (training-d, $n = 8$; training-f, $n = 17$) carry very little weight, and their figures should be read as tendencies rather than findings. training-c ($n = 4$) contains a single class in the test split and is omitted.

D. Isolating the site shortcut

The per-site breakdown shows that performance does not transfer, but it does not by itself prove that the site is what the features encode. We test that directly, on the training split only so that the test partition of Table III stays sealed, with three grouped cross-validation experiments driven by one fixed balanced random forest (Table VI, Fig. 8).

First, the site is plainly recoverable. A classifier asked to name the sub-database from the MFCC features alone reaches a balanced accuracy of 0.647 against the 0.167 chance rate of the six-site problem (0.549 on PWP), well above chance on the metric this paper is defined by (Table VI, block A). We deliberately quote the balanced figure rather than raw accuracy: the site distribution is heavily skewed — training-e alone is roughly two thirds of the recordings — so a predictor that always names the majority site would already score about 66%, and raw accuracy against a 0.167 reference would badly overstate recoverability. The substrate for a shortcut is

TABLE VI

ISOLATING THE SITE SHORTCUT. ALL THREE BLOCKS ARE COMPUTED ON THE *training* PARTITION BY GROUPED CROSS-VALIDATION; THE TEST SPLIT OF TABLE III IS NEVER OPENED HERE. A SINGLE FIXED, BALANCED RANDOM FOREST IS USED THROUGHOUT, BECAUSE THE QUANTITY OF INTEREST IS THE GAP BETWEEN EVALUATION PROTOCOLS RATHER THAN ANY ABSOLUTE SCORE. SITE RECOVERABILITY IS REPORTED AS MODIFIED ACCURACY (BALANCED OVER THE SIX SITES, M_{ACC} CHANCE 0.167) RATHER THAN RAW ACCURACY, SINCE THE SITE DISTRIBUTION IS HEAVILY SKEWED AND RAW ACCURACY WOULD BE INFLATED BY THE MAJORITY SITE; DIAGNOSIS M_{ACC} CHANCE IS 0.500.

	MFCC	PWP
<i>A. Is the site recoverable from the features?</i>		
Site M _{ACC} (balanced over sites)	0.647	0.549
<i>B. Does the diagnosis transfer across sites?</i>		
Diagnosis M _{ACC} , pooled CV	0.840	0.810
Diagnosis M _{ACC} , leave-one-site-out	0.457	0.472
Drop (pooled – LOSO)	0.382	0.338
<i>C. Does signal survive once site is fixed?</i>		
Diagnosis M _{ACC} , within-site	0.756	0.703

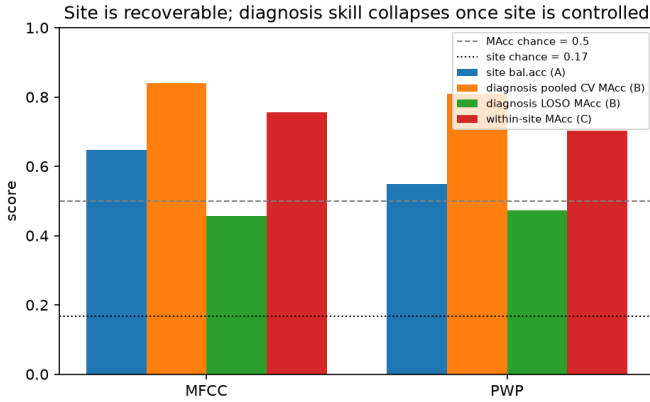


Fig. 8. Isolating the site shortcut on the training partition. Recording site is recoverable from the features well above chance (blue); pooled cross-validated diagnosis M_{ACC} (orange) collapses to chance under leave-one-site-out evaluation (green); a weak diagnosis signal nonetheless survives once site is held fixed (red). Test data is not used.

nonetheless plainly present in the representation itself, before any diagnostic label is introduced.

Second, the diagnosis does not transfer across that structure. When the same features are used for the intended task but the model is evaluated *leave-one-site-out* — trained on five sites and tested on the held-out sixth — M_{ACC} falls from 0.840 pooled to 0.457 on MFCC, and from 0.810 to 0.472 on PWP. Both land at the 0.500 of a constant predictor. The pooled figure that anchors the leaderboard describes performance *within* the corpus’s site mixture; almost none of it survives a change of acquisition site, which is the condition a screening deployment would actually face.

Third — and this is the qualification that keeps the conclusion honest — the features are not *only* a site fingerprint. Held at a fixed site, where prevalence can no longer stand in for the label, diagnosis M_{ACC} is 0.756 on MFCC and 0.703 on PWP, clearly above chance. That average is, however, inflated by

TABLE VII

CONTEXT FROM PUBLISHED PHYSIONET/CINC 2016 ENTRIES. THESE WERE SCORED ON THE OFFICIAL HIDDEN TEST SET, WHICH WAS NEVER RELEASED; OUR NUMBERS COME FROM A HELD-OUT SPLIT OF THE PUBLIC TRAINING DATA. THE COMPARISON IS THEREFORE CONTEXTUAL AND *not* LIKE-FOR-LIKE.

System	M _{ACC}	Se.	Sp.	Same test set
This work (SVM–MFCC)	0.866	0.950	0.782	–
Potes et al. [6]	0.860	0.942	0.778	✓
Zabihi et al. [8]	0.859	0.869	0.849	✓
Rubin et al. [7]	0.839	0.727	0.951	✓

two untrustworthy partitions: *training-c* ($n = 22$) reports a within-site M_{ACC} of 1.000 and *training-e* (abnormal prevalence 7.4 %) of 0.985, both artefacts of tiny or extremely skewed samples. On the three large, better-balanced sites the within-site M_{ACC} is a more sober 0.55–0.64. The reading we take from all three blocks is that the pooled metric is substantially, but not entirely, a site-prevalence artefact: a genuine but weak diagnostic signal exists, and the aggregate score overstates it by roughly the 0.34–0.38 that evaporates under leave-one-site-out evaluation.

E. Overfitting and calibration

Two weaknesses are worth stating. Both random forests show a large gap between training and cross-validated balanced accuracy — 0.947 versus 0.867 on MFCC, and 0.945 versus 0.814 on PWP — so they are memorising the training folds despite the depth and leaf-size grid; stronger regularisation would be the obvious next step. Second, the classical models are poorly calibrated, with expected calibration error 0.129 for RF–MFCC and 0.150 for RF–PWP. Since recording-level decisions are formed by averaging window probabilities, miscalibration propagates into the aggregation, and the close agreement between mean-probability and majority-vote aggregation (within 0.004 M_{ACC}) suggests the probabilities are contributing little beyond their ordering.

F. Comparison with published work

Table VII places our best result alongside published CinC 2016 entries. Our M_{ACC} of 0.866 sits at the top of the 0.84–0.86 band those entries occupy. This must not be read as an improvement. The published figures were obtained on the official hidden test set, which was never released; ours come from a self-defined split of the public training data, with a different class balance, a different site mix and a different difficulty. The numbers are of the same order, which is the only conclusion the comparison supports, and it is reassuring mainly as evidence that the pipeline is not broken.

VI. RESULTS III: EXPLAINABILITY AS A TEST

Displaying a few heat maps and asserting that a model “focuses on clinically relevant regions” is unfalsifiable as stated. This section instead states two predictions in advance and tests each with the tool suited to it.

Two tools, two axes. Our explainability evidence comes from two methods that describe different things, and keeping

TABLE VIII

WHAT EACH EXPLAINABILITY EXPERIMENT TESTS. “TARGET” IS THE AXIS THE METHOD IS READ ON; THE OTHER AXIS IS NOT CLAIMED.

Experiment	Model	Method	Target
Weight randomisation	CNN	Grad-CAM	(validity)
Energy confound	CNN	Grad-CAM	(validity)
Variant robustness	CNN	Grad-CAM vs. ++	(validity)
Frequency profile	SVM, RF	SHAP	frequency
Cross-model agreement	SVM, RF	SHAP	frequency
Cardiac-cycle enrichment	CNN	Grad-CAM	time
Outcome stratification	CNN	Grad-CAM	time

them apart is essential to reading the results. SHAP is applied to the *classical* models (SVM and random forest, on MFCC and PWP features) and answers a question about *frequency*: which spectral regions carry the attribution. Grad-CAM is applied to the *CNN* (on the log-Mel spectrogram) and answers a question about *time*: where in the cardiac cycle the attribution falls. The two are orthogonal by construction, and deliberately so. SHAP acts on pooled feature vectors, from which the time axis has already been discarded, so it can speak only to frequency; the trustworthiness of that frequency mapping is why it is used there. Grad-CAM produces a two-dimensional time–frequency map, but its last convolutional feature map is only a few cells wide, so its frequency axis is too coarse for spectral claims while its time axis (188 frames) is fine-grained. We therefore read each method on the axis where it is reliable and on no other. There is consequently no common axis on which a SHAP frequency profile and a Grad-CAM temporal profile could be compared directly, and we do not attempt such a comparison; the cross-method agreement we *can* measure is agreement among independent SHAP explainers, reported in Section VI-B.

The specific estimators: for SHAP, the exact tree estimator on the random forests and the model-agnostic kernel estimator on the SVMs [16], [19]; for the CNN, Grad-CAM at the last convolutional block [20]. Table VIII summarises which model and method each experiment below uses and which axis it is allowed to speak to.

A. Sanity checks: is the CNN’s Grad-CAM meaningful at all?

Model: CNN. Method: Grad-CAM. Tested: whether the attribution reflects learned weights rather than architecture or loudness. These checks are reported before any finding, because an attribution map means nothing until it is shown to depend on the trained weights [21].

Weight randomisation. Recomputing Grad-CAM from a randomly re-initialised copy of the same architecture gives maps that correlate with the trained model’s at only 0.051 over 100 windows. The maps therefore reflect what the network learned, not its wiring. *Pass.*

Energy confound. Correlating the CNN’s temporal attribution profile with the signal envelope gives 0.068 over 499 windows, with no window above 0.5. Attribution is not merely tracking loudness — which matters because S1 and S2 are the

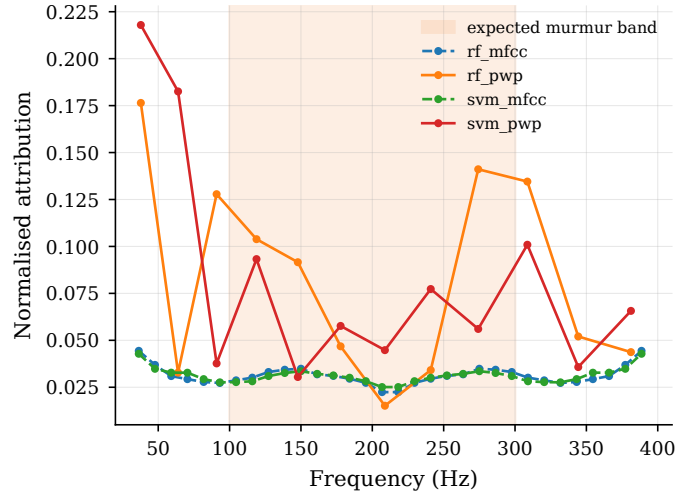


Fig. 9. SHAP attribution mapped to frequency for the four classical feature/model pairings. PWP features map onto bands exactly; the MFCC profile is obtained by redistributing attribution through the magnitude of the DCT basis and is approximate.

loud events, so a loudness-driven map would confound the timing claim of Section VI-C. *Pass.*

Variant robustness. Grad-CAM and Grad-CAM++ [22] maps correlate at only 0.392. The temporal conclusions below are therefore specific to the Grad-CAM configuration and should not be assumed to transfer to other CAM variants. *Reported as a limitation.*

B. Where in frequency? (SHAP, classical models)

Models: SVM and RF, on MFCC and PWP. Method: SHAP. Tested: which frequencies drive the classical models, and whether independent explainers agree on the answer. This is the only place the frequency axis is claimed, and Grad-CAM contributes nothing to it.

The frequency mapping is exact for PWP and approximate for MFCC, and the distinction governs which number to trust. A PWP feature belongs to exactly one frequency band by construction, so its SHAP value maps onto frequency directly. An MFCC coefficient is a weighted sum over all mel bands, so its attribution must be redistributed through the magnitude of the DCT basis; that projection is sign-agnostic and therefore approximate. Where the two disagree, the PWP result is the reliable one.

Cross-model agreement is weak. All four classical pairings peak at the low band edge, 25 Hz (Fig. 9), but the mean pairwise correlation between the four frequency profiles is only 0.514, and that average hides a wide range. The two MFCC models agree almost perfectly with each other (0.961), whereas RF–PWP and SVM–MFCC agree at 0.280. The high figure is a statement about a shared *representation*, not a shared physiological finding: models built on the same features attribute similarly, models built on different features do not. Agreement across independent explainers would have been the strongest single piece of interpretability evidence available — it is what a genuine cross-method check looks like on

TABLE IX
CARDIAC-CYCLE ALIGNMENT OF THE CNN’S GRAD-CAM ATTRIBUTION.
 E_s IS ENRICHMENT, ATTRIBUTION MASS DIVIDED BY THE TIME
FRACTION OCCUPIED BY STATE s ; $E_s = 1$ IS EXACTLY WHAT UNIFORM
TEMPORAL ATTENTION PRODUCES. COMPUTED ON THE 66.5% OF TEST
WINDOWS (2235 OF 3362) THAT THE SEGMENTER LABELLED WITH
SUFFICIENT CONFIDENCE.

Cardiac state	Attribution mass	E_s
S1	0.190	1.001
systole	0.245	1.056
S2	0.150	0.753
diastole	0.415	1.067

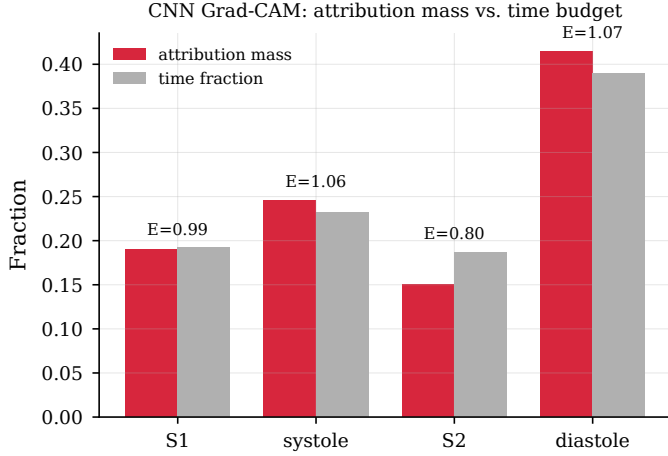


Fig. 10. Distribution of per-window systolic enrichment E_{systole} for the CNN. The dashed line at $E = 1$ is what uniform temporal attention produces; the distribution sits slightly to its right.

the frequency axis — and it did not materialise. We report frequency agreement as weak.

The individual PWP features are plausible but not decisive. The top two by mean absolute SHAP value are the Shannon entropy of band 0 (25–51 Hz, 9.5% of total attribution) and of band 9 (291–326 Hz, 7.9%), followed by relative energy in 77–105 Hz. Spectral entropy in a low band and a high band is a sensible murmur descriptor, since murmurs are noise-like where S1 and S2 are tonal. But their sign consistency is only 0.57 and 0.55: they push toward “abnormal” barely more often than toward “normal”. We therefore describe them as the features the model relies on most, not as evidence of a murmur detector. For the MFCC models the leading features are MFCC₃ mean and MFCC₀ statistics; the prominence of the zeroth coefficient, which encodes overall energy, is not encouraging given the site dependence of Section V-C.

C. Where in the cardiac cycle? (Grad-CAM, CNN)

Model: CNN. Method: Grad-CAM. Tested: whether the CNN’s attention concentrates in systole, where clinically important adult murmurs occur. This is the only place the time axis is claimed, and SHAP contributes nothing to it.

An envelope-based segmenter, operating independently of the CNN, labels each window’s timeline into S1, systole,

S2 and diastole. Given a unit-mass Grad-CAM map, the enrichment for cardiac state s is

$$E_s = \frac{\text{fraction of attribution mass inside } s}{\text{fraction of time occupied by } s}, \quad (1)$$

so $E_s = 1$ is exactly what uniform temporal attention produces, $E_s > 1$ is a genuine preference, and $E_s < 1$ is avoidance. The normalisation is the point: “25% of attention fell in systole” is uninterpretable until one knows systole occupies 23% of the window. Because most clinically important adult murmurs are systolic, a CNN that has learned pathology should show $E_{\text{systole}} > 1$ — a prediction fixed before the measurement.

Confirmed in sign, negligible in magnitude. Table IX gives $E_{\text{systole}} = 1.056$, above the uniform null by about 6%. Against a permutation null — the same attribution profile shuffled in time, so its sparsity and dynamic range survive but any relation to the cardiac cycle is destroyed — only 33% of the 200 tested windows reach $\alpha = 0.05$, the mean p -value is 0.43, and the effect size is Cohen’s $d = 0.131$ (below the conventional threshold for a small effect). Fig. 10 shows a distribution shifted right of $E = 1$ but overlapping it heavily. This is a tendency, not a property of the model.

Three conditions attach to these numbers. First, the segmenter is a simplified envelope-based labeller, not the reference hidden semi-Markov model [23]; it is used only at evaluation and never in training, so a noisier segmenter adds variance and makes the test *harder* to pass rather than easier, and cannot manufacture a positive result. Second, low-confidence windows are excluded, so the honest denominator for every figure here is the 66.5% inclusion rate (2235 of 3362 windows, mean segmenter confidence 0.559, mean estimated heart rate 82 bpm). Third, as stated at the outset, Grad-CAM’s coarse frequency axis means no spectral claim is drawn from it; the frequency evidence is entirely SHAP’s.

D. The check that decides it: right for the right reason? (Grad-CAM, CNN)

Model: CNN. Method: Grad-CAM. Tested: whether the systolic tendency distinguishes correct from incorrect decisions — the difference between the CNN attending to systole and the CNN using systolic evidence.

If the systolic tendency reflected murmur detection, it should be stronger when the CNN is right than when it is wrong. Stratifying the same Grad-CAM enrichment by outcome, it is not: mean E_{systole} is 1.085 on true positives but 1.115 on false positives and 1.157 on false negatives ($n = 26$, so the last is a tendency at best). The true-minus-false-positive difference is -0.030 — the systolic focus is marginally *stronger* when the CNN raises a false alarm.

This is the clearest negative result in the project, and it bounds what the rest of the section may claim. The maps are not artifacts — the sanity checks establish that — and they do lean toward systole. But the lean does not separate correct from incorrect decisions, so it cannot be evidence that the CNN uses systolic information to decide. It behaves more

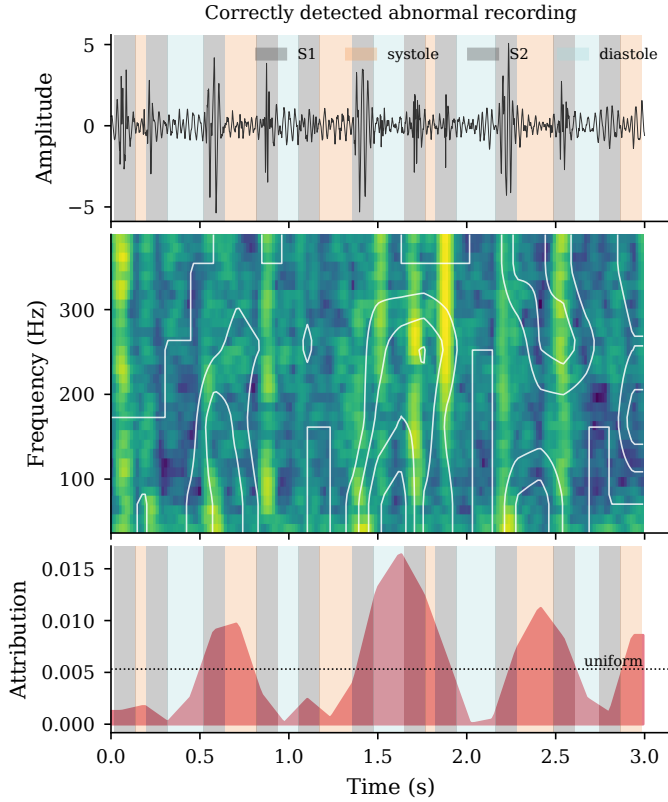


Fig. 11. Grad-CAM on a correctly classified abnormal window, with the segmenter state track below. Maps such as this motivate the enrichment test but cannot, on their own, support it.

like a fixed positional habit of the architecture than like a learned mechanism. Fig. 11 shows a representative correctly classified abnormal window: visual inspection of such a map is what suggested the hypothesis, and the quantitative test is what stops us concluding it.

VII. DISCUSSION AND LIMITATIONS

Three diagnostics were applied to a system whose aggregate score, MAcc 0.866, sits comfortably within the published range for this corpus, and each isolates a different weakness that the headline number hides.

Unsupervised structure (k-means on the feature spaces). In two of three feature spaces, *k*-means recovers the acquisition site far better than the diagnosis, so the class boundary is a thin supervised direction inside a geometry organised by how and where the recording was made.

Per-site performance (best model, SVM-MFCC). MAcc ranges from 0.986 to 0.477 across sub-databases, with the pooled figure carried by the single largest and easiest one; performance does not transfer across acquisition conditions.

Attribution (Grad-CAM on the CNN). The CNN’s attention is genuinely weight-dependent and mildly systole-preferring, but no more so on correct decisions than on incorrect ones, so the tendency cannot be read as use of systolic evidence.

None of the three establishes a mechanism on its own, so we ran the experiment that ties them together (Section V-D):

predicting the sub-database directly from the features, and re-evaluating the diagnosis across sites rather than pooled. It confirms the inference, with one qualification. The site is recoverable at 0.647 balanced accuracy against 0.167 chance, so the shortcut substrate is real; and pooled diagnosis MAcc collapses from 0.840 to 0.457 under leave-one-site-out evaluation, so almost none of the headline performance transfers across acquisition conditions. The qualification is that a within-site signal does not vanish: held at a fixed site, diagnosis MAcc is 0.70–0.76, and even after discounting two tiny or extreme-prevalence sites that inflate that average, the large sites retain a modest 0.55–0.64. The features are therefore not *only* a site fingerprint. The defensible summary is the measured one: the aggregate metric is substantially, not entirely, a site-prevalence artefact, and it overstates the transferable diagnostic signal by roughly the 0.34–0.38 that disappears when the site is no longer shared between training and test.

The two-tool design of Section VI is a limitation as much as a feature. Because SHAP was read only on frequency and Grad-CAM only on time, the study never obtains a single explanation spanning both axes, and the one genuinely cross-method check available — agreement among the four independent SHAP frequency profiles — came out weak (0.514). A time–frequency attribution that could be trusted on both axes at once, for instance a higher-resolution CAM or an input-space gradient method on the spectrogram, would allow the frequency and timing claims to be tested jointly rather than separately; we leave that to future work.

Several further limitations should be stated plainly. The evaluation rests on one split of one corpus; recording-level splitting and joint stratification guard against the most common leakage, but a single split cannot separate representation effects from split-specific variance, and repeated splits or nested cross-validation would give firmer footing. Both random forests overfit their training folds and neither classical model is well calibrated. The MFCC-to-frequency projection used for SHAP is sign-agnostic and approximate, and disagrees with the exact PWP mapping. The CNN’s Grad-CAM and Grad-CAM++ maps agree at only 0.392, so the temporal result is specific to one variant. The segmenter labels only 66.5 % of test windows with sufficient confidence, and nothing is known about whether the excluded third differs systematically — if the low-confidence windows are the noisy ones, they may also be the ones where site cues matter most. Finally, the two smallest sub-databases contribute 8 and 17 test recordings and carry essentially no statistical weight.

We also note what did work. All five models substantially exceed trivial baselines, and the ranking of feature domains is stable and interpretable: MFCC outperforms PWP under both classifier families, and the log-Mel CNN attains the best ranking quality (AUC 0.961) despite an operating point that gives it no MAcc advantage. The comparison the study was designed to make is answered, even though the answer is not the one anticipated.

VIII. CONCLUSION

This project implemented a heart sound classification pipeline on the public portion of PhysioNet/CinC 2016 and compared three feature representations across three classifier families under a single recording-level protocol. The best configuration, an RBF SVM on MFCC statistics, reaches a modified accuracy of 0.866 on our held-out split, and the log-Mel CNN is statistically indistinguishable from it on that metric while making significantly fewer total recording-level errors.

The more useful findings are the ones that came from testing those numbers rather than reporting them. k -means on the unsupervised feature spaces recovers the recording site better than the diagnosis; per-site modified accuracy spans 0.509, from near-perfect on the largest sub-database to below chance level on two others; and the CNN's attribution, though verifiably dependent on learned weights and mildly enriched in systole ($E_{\text{systole}} = 1.056$), is not more systole-enriched on correct predictions than on incorrect ones. Two of the three expectations we set out with were contradicted and one was only weakly supported.

Making site an explicit variable settled the central question. A classifier predicts the sub-database from the same features at 0.647 balanced accuracy (0.167 chance), and evaluating the diagnosis leave-one-site-out rather than pooled drops MAcc from 0.840 to 0.457, close to chance — while a weak diagnostic signal (0.55–0.64 on the large sites) survives once site is held fixed. On this corpus, then, the aggregate metric substantially overstates what the models have learned, though it does not reduce entirely to a site fingerprint, and the diagnostics — not the leaderboard position — are where the informative content of such a study lies.

The continuation this points to is no longer a measurement but a remedy: training that is explicitly invariant to site, for instance a domain-adversarial or group-distributionally-robust objective, evaluated leave-one-site-out and, ideally, validated on an independent corpus. That is the setting in which a heart sound classifier would have to demonstrate that it has learned cardiac acoustics rather than the room they were recorded in.

REPRODUCIBILITY

The pipeline is driven by a single YAML configuration whose hash is recorded in every result artifact, and every number in this report is read programmatically from those artifacts rather than transcribed by hand. Random seeds are fixed and propagated to NumPy, scikit-learn and PyTorch.

REFERENCES

- [1] M. A. Chizner, "Cardiac auscultation: Rediscovering the lost art," *Current Problems in Cardiology*, vol. 33, no. 7, pp. 326–408, 2008.
- [2] A. Leatham, *Auscultation of the Heart and Phonocardiography*, 2nd ed. Churchill Livingstone, 1975.
- [3] A. K. Abbas and R. Bassam, "Phonocardiography signal processing," *Synthesis Lectures on Biomedical Engineering*, vol. 4, no. 1, pp. 1–194, 2009, source for the spectral content of S1, S2 and murmurs.
- [4] C. Liu, D. Springer, Q. Li, B. Moody, R. A. Juan, F. J. Chorro, F. Castells, J. M. Roig, I. Silva, A. E. W. Johnson, Z. Syed, S. E. Schmidt, C. D. Papadaniil, L. Hadjileontiadis, H. Naseri, A. Moukadem, A. Dieterlen, C. Brandt, H. Tang, M. Samieinasab, M. R. Samieinasab, R. Sameni, R. G. Mark, and G. D. Clifford, "An open access database for the evaluation of heart sound algorithms," *Physiological Measurement*, vol. 37, no. 12, pp. 2181–2213, 2016.
- [5] G. D. Clifford, C. Liu, B. Moody, D. Springer, I. Silva, Q. Li, and R. G. Mark, "Classification of normal/abnormal heart sound recordings: The physionet/computing in cardiology challenge 2016," *Computing in Cardiology*, vol. 43, pp. 609–612, 2016.
- [6] C. Potes, S. Parvaneh, A. Rahman, and B. Conroy, "Ensemble of feature-based and deep learning-based classifiers for detection of abnormal heart sounds," in *Computing in Cardiology Conference (CinC)*, 2016, pp. 621–624, challenge winner, MAcc 0.860.
- [7] J. Rubin, R. Abreu, A. Ganguli, S. Nelaturi, I. Matei, and K. Sricharan, "Classifying heart sound recordings using deep convolutional neural networks and mel-frequency cepstral coefficients," in *Computing in Cardiology Conference (CinC)*, 2016, pp. 813–816, mAcc 0.839.
- [8] M. Zabihi, A. B. Rad, S. Kiranyaz, M. Gabbouj, and A. K. Katsaggelos, "Heart sound anomaly and quality detection using ensemble of neural networks without segmentation," in *Computing in Cardiology Conference (CinC)*, 2016, pp. 613–616, mAcc 0.859; notable for avoiding segmentation entirely.
- [9] F. Safara, S. Doraisamy, A. Azman, A. Jantan, and A. R. Abdullah Ramaiah, "Multi-level basis selection of wavelet packet decomposition tree for heart sound classification," *Computers in Biology and Medicine*, vol. 43, no. 10, pp. 1407–1414, 2013, perceptual wavelet packet features for PCG.
- [10] R. R. Coifman and M. V. Wickerhauser, "Entropy-based algorithms for best basis selection," *IEEE Transactions on Information Theory*, vol. 38, no. 2, pp. 713–718, 1992.
- [11] S. Mallat, *A Wavelet Tour of Signal Processing: The Sparse Way*, 3rd ed. Academic Press, 2008.
- [12] A. J. DeGrave, J. D. Janizek, and S.-I. Lee, "Ai for radiographic covid-19 detection selects shortcuts over signal," *Nature Machine Intelligence*, vol. 3, no. 7, pp. 610–619, 2021, motivates our per-site shortcut analysis.
- [13] S. E. Schmidt, C. Holst-Hansen, C. Graff, E. Toft, and J. J. Struijk, "Segmentation of heart sound recordings by a duration-dependent hidden markov model," *Physiological Measurement*, vol. 31, no. 4, pp. 513–529, 2010, source of the spike-removal procedure used in preprocessing.
- [14] S. Davis and P. Mermelstein, "Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 28, no. 4, pp. 357–366, 1980, original MFCC formulation.
- [15] I. Daubechies, "Orthonormal bases of compactly supported wavelets," *Communications on Pure and Applied Mathematics*, vol. 41, no. 7, pp. 909–996, 1988.
- [16] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee, "From local explanations to global understanding with explainable ai for trees," *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020, treeSHAP.
- [17] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, "Physiobank, physiotoolkit, and physionet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.
- [18] T. G. Dietterich, "Approximate statistical tests for comparing supervised classification learning algorithms," *Neural Computation*, vol. 10, no. 7, pp. 1895–1923, 1998, justifies McNemar's test for paired classifier comparison.
- [19] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4765–4774.
- [20] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in *IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.
- [21] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, model-randomisation test used in our phase 08.
- [22] A. Chattopadhyay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-cam++: Generalized gradient-based visual explanations for deep

convolutional networks,” in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2018, pp. 839–847.

- [23] D. B. Springer, L. Tarassenko, and G. D. Clifford, “Logistic regression-hsmm-based heart sound segmentation,” *IEEE Transactions on Biomedical Engineering*, vol. 63, no. 4, pp. 822–832, 2016.